

# JULIO CÉSAR GALINDO LÓPEZ

## Curriculum Vitae

Mexico City, Mexico | jcgalingo@up.edu.mx | Universidad Panamericana, Ciudad UP

### RESEARCH PROFILE

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Research at the intersection of algebraic topology and algebraic geometry, with particular emphasis on Chow groups and motivic methods. Additional work focuses on mathematical models for finance and economics, especially machine learning, artificial intelligence, stochastic simulation, and data-driven prediction of markets, consumer behavior, and investment strategies. More than 12 years of university teaching experience across UNAM, Universidad Panamericana, Tecnológico de Monterrey CCM, and BUAP.

### RESEARCH INTERESTS

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Algebraic Geometry; Algebraic Topology; Statistics; Probability; Stochastic Processes; Stochastic Calculus; Quantitative Finance; Economic Models; Time Series; Stochastic Simulation; Programming; Data Management; Quantum Computing; Data Science; Machine Learning; Artificial Intelligence.

### EDUCATION

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| 2016–2021 | <b>Ph.D. in Mathematical Sciences, National Autonomous University of Mexico (UNAM)</b><br><i>Instituto de Matemáticas, Mexico City</i><br>Dissertation: Algebraic Cycles via Motivic Homotopy Theory: On the Convergence of the Orthogonal Tower. Advisor: Pablo Peláez Menaldo - Senior Researcher, Chebyshev Laboratory, St. Petersburg State University. |
| 2014–2015 | <b>M.Sc. in Mathematical Sciences, National Autonomous University of Mexico (UNAM)</b><br><i>Instituto de Matemáticas, Cuernavaca</i><br>Thesis: Drinfeld Reciprocity Law. Advisor: Timothy Mooney Gendron Thornton.                                                                                                                                        |
| 2008–2013 | <b>B.S. in Actuarial Science, National Autonomous University of Mexico (UNAM)</b><br><i>Facultad de Ciencias, Mexico City</i><br>Thesis: Random Matrices and L-Functions. Advisor: Eduardo Dueñez Guzmán. Co-advisor: Víctor Manuel Pérez-Abreu Carrión.                                                                                                    |

### MITX AND PROFESSIONAL TRAINING

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| 2022 | <b>MITx MicroMasters Program in Statistics and Data Science</b><br><i>Massachusetts Institute of Technology</i><br>Specializations completed: Machine Learning with Python: From Linear Models to Deep Learning; Probability — The Science of Uncertainty and Data. |
| 2022 | <b>Additional Professional Training</b> <ul style="list-style-type: none"><li>• Python Application using PyMongo and MongoDB Database (Project Network).</li><li>• Microsoft Office Specialist.</li></ul>                                                           |

### ACADEMIC APPOINTMENTS

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| Present      | Full-Time Research Professor, Universidad Panamericana, Ciudad UP                                                                                                                                                                                                             |
| 2023–2024    | <b>Research Professor (Associate Level C), Benemérita Universidad Autónoma de Puebla (BUAP)</b><br><i>Faculty of Physics and Mathematics</i><br>Teaching and research in business consulting, financial risk management, portfolio theory, and Monte Carlo methods in Python. |
| 2023–present | <b>Lecturer, IIMAS–FES Acatlán, UNAM</b><br><a href="https://www.up.edu.mx/campus-ciudad-panamericana/">https://www.up.edu.mx/campus-ciudad-panamericana/</a>                                                                                                                 |

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| 2016–present | <p>Courses include Statistical Computing and Probability &amp; Stochastic Simulation.</p> <p><b>Lecturer, Applied Mathematics and Computing, FES Acatlán, UNAM</b></p> <p>Courses include Probability, Calculus III–IV, Logic, Geometry, and Numerical Methods II.</p> |
| 2021–2023    | <p><b>Adjunct Professor, Tecnológico de Monterrey CCM</b></p> <p>Courses in Software Development, Data Analytics and AI Tools, Data Science, Data Visualization, Technology Services Strategy, and Natural Language Processing.</p>                                    |
| 2011–2022    | <p><b>Lecturer, Faculty of Sciences, UNAM</b></p> <p>Courses include Probability I–II, Stochastic Processes I–II, Mathematical Analysis I, Measure Theory, Algebra, Calculus IV, and seminars in actuarial applications and arithmetic geometry.</p>                   |
| 2020         | <p><b>Adjunct Professor, Universidad Panamericana (Upper Secondary Division)</b></p> <p>Computer Science: Python for Data Science; Introduction to Python for Data Science.</p>                                                                                        |
| 2016–2017    | <p><b>Graduate Teaching Assistant, Universidad Autónoma Metropolitana</b></p> <p>Integral Calculus and Linear Algebra.</p>                                                                                                                                             |

## SELECTED PUBLICATIONS

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| 2016           | Galindo, J. C. La ley de reciprocidad de Drinfeld. <i>Morfismos Cinvestav</i> 20(2): 1–66.                                                                                                                                              |
| 2019           | Galindo, J. C. Convergence of the birational tower spectral sequence for mixed motives. <i>Homology, Homotopy and Applications</i> 16(2): 227–301.                                                                                      |
| 2022           | Galindo, J. C. On the convergence of the orthogonal spectral sequence and applications. <i>Homology, Homotopy and Applications</i> 24(2): 389–401.                                                                                      |
| 2022           | Galindo, J. C. On the Betti realization for the birational tower and applications. Listed in the CV as forthcoming in <i>Homology, Homotopy and Applications</i> .                                                                      |
| 2022           | Galindo, J. C. The orthogonal tower and the E1-term of Voevodsky’s slice spectral sequence. Listed in the CV as under review.                                                                                                           |
| 2024           | Additional works in progress listed in the CV include A Quantum Categorical Approach to Portfolio Theory; Deep Learning and Random Matrices: A Case Study from INEGI Data; and An Approximation to the Adams-Novikov Spectral Sequence. |
| 2026           | Galindo, J. C., and Marisol Velázquez-Salazar. Reconfiguration of Global Value Chains in Mexican Coffee: Labor, Ethics, and Sustainability from a Humanistic Perspective. In preparation.                                               |
| In preparation | Galindo, J. C. The Orthogonal Spectral Sequence and Higher Chow Groups. In preparation.                                                                                                                                                 |
| In preparation | Galindo, J. C. Cover times for rare gap events in Dyson Brownian motion: Poisson point processes and Gumbel fluctuations. In preparation.                                                                                               |

## CONSULTING AND APPLIED EXPERIENCE

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Risk consulting work has included actuarial modeling for financial and insurance risk assessment using R, Python, SAS, and Julia; historical-data analysis for the Comisión Nacional de Seguros y Fianzas; and strategic consulting for clients such as Carl Zeiss on insurance portfolio optimization, risk mitigation, pricing, and actuarial-process automation.

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| 2023–2024 | <b>Carl Zeiss Group — business-model management using blockchain.</b>                                                            |
| 2023–2024 | <b>Comisión Nacional de Seguros y Fianzas — premium forecasting via time-series methods.</b>                                     |
| 2023–2024 | <b>Comisión Nacional de Seguros y Fianzas — gender-related policy analysis using sentiment analysis and boosting techniques.</b> |
| 2023–2024 | <b>Review of Calculus I and Calculus IV curricula for Applied Mathematics and Computing, FES Acatlán.</b>                        |

2023–2024 UNAM Diploma Program in Statistics and Data Mining — Module II: Probability and Stochastic Simulation.

## HONORS, SERVICE, AND ACADEMIC ACTIVITIES

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2024 Recognition for Academic Service, FES Acatlán, UNAM.  
2024– Candidate, Mexican National System of Researchers (SNI), CONAHCyT.  
2019–2023 Doctoral scholarship, CONACyT.  
2016–2019 Master’s scholarship, CONACyT.  
2024 UNAM recognition for ten years of academic service.  
2023 Referee for Advances in Mathematics.  
2023 Working group for the planning of the M.Sc. in Data Science Research, BUAP.  
2009–2011 Research Assistant within Mexico’s National System of Researchers (SNI III), CIMAT.

## SELECTED CONFERENCE PARTICIPATION AND PRESENTATIONS

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2026 “Reconfiguration of Global Value Chains in Mexican Coffee: Labor, Ethics, and Sustainability from a Humanistic Perspective,” with Marisol Velázquez-Salazar. WSSA, March 26, 2026.  
  
2019 52nd Congress of the Mexican Mathematical Society, Universidad de Nuevo León.  
2018 Motives in St. Petersburg, Euler International Mathematical Institute, St. Petersburg, Russia.  
  
2016 CIMPA–CIMAT–ICTP Summer School on Moduli of Curves, CIMAT, Guanajuato.  
2016 National Congress of Algebraic Geometry, Casa Matemática Oaxaca.  
2014 Applications of Étale Cohomology Summer School, Alfréd Rényi Institute of Mathematics, Budapest.  
  
2011 35th Conference on Stochastic Processes and their Applications (SPA), Oaxaca, Mexico.  
  
2010 “Random Matrices and the Riemann Zeta Function,” Second Interinstitutional Seminar on Random Matrices, CIMAT.  
  
2011 Mini-course: “Random Matrices and Number Theory,” Third Interinstitutional Seminar on Random Matrices, CIMAT.

## STUDENT SUPERVISION

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2024 M.Sc. advisor for José Alberto Meléndez Piña (IMATE–UNAM): Fourier–Mukai Transform and Computation in Derived Categories.  
2024 Undergraduate advisor for Francisco Estrada Ruano (FES Acatlán, UNAM): Random Matrices and Determinantal Stochastic Processes (in progress).  
2023 Undergraduate advisor for Fernando Flores Tiburcio (Faculty of Sciences, UNAM): Almost Coherent Sheaves on Compact Riemann Surfaces (honorable mention).  
2024 Committee member for Martín Guarneros Martínez: p-adic Differential Calculus.  
2014 Committee member for Eduardo Granados González: Development of a Random Number Generator Based on an Air Lottery Machine.

## LANGUAGES AND TECHNICAL SKILLS

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**Languages** Spanish (native); English (intermediate); French (intermediate); Russian (intermediate).  
**Technical skills** R, Python, Matlab, Scilab, SQL, Mathematica, Java, C, C++, C#, LaTeX, HTML.